北京交通大學

信号与系统

School of Electronic
and Information Engineering

Zhou Hang

2025.4

BJTU Signals and Systems Course Group, Copyright© 2025

Frequency domain analysis of signals

- Frequency domain analysis of continuous periodic signals
- Frequency domain analysis of continuous aperiodic signals
- Frequency domain analysis of discrete periodic signals
- Frequency domain analysis of discrete aperiodic signals
- Time domain sampling of continuous-time signals
- Analysis of signal spectrum by MATLAB

Frequency domain analysis of discrete aperiodic signals

- ※ Frequency domain representation of discrete aperiodic signals
- ※ Spectrum of discrete aperiodic signals
- ※ Properties of discrete aperiodic sequence DTFT

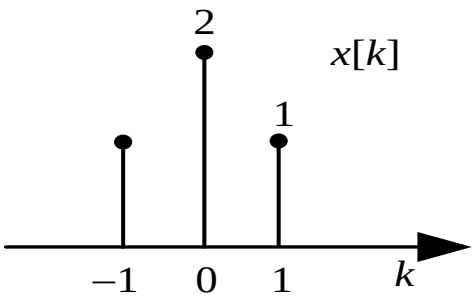
Frequency domain representation of discrete aperiodic signals

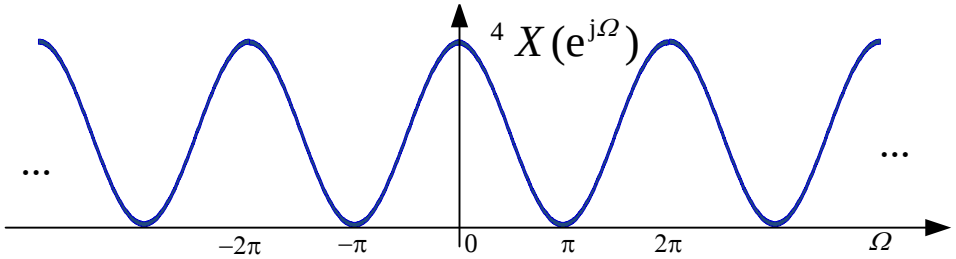
Aperiodic sequences $x[k]$ satisfying certain convergence conditions

can employ the imaginary exponential sequence to be expressed as:

$$x[k]=\frac{1}{2\pi}\int_{<2\pi>} X(\mathrm{e}^{\mathrm{j}\Omega})\mathrm{e}^{\mathrm{j}\Omega k}\mathrm{d}\Omega$$

The weighting coefficient $X(\mathrm{e}^{\mathrm{j}\Omega})$ is called the spectrum of discrete aperiodic signals $x[k]$.





Frequency domain representation of discrete aperiodic signals

IDTFT $$x[k]=\frac{1}{2\pi}\int_{<2\pi>}X(\mathrm{e}^{\mathrm{j}\Omega})\mathrm{e}^{\mathrm{j}\Omega k}\mathrm{d}\Omega$$

DTFT $$X(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=-\infty}^{\infty}x[k]\mathrm{e}^{-\mathrm{j}\Omega k}$$

$X(\mathrm{e}^{\mathrm{j}\Omega})$ is called the spectrum of discrete aperiodic signals $x[k]$.

$$x[k]\xleftrightarrow{\text{DTFT}}X(\mathrm{e}^{\mathrm{j}\Omega})$$

Frequency domain representation of discrete aperiodic signals

Convergence of DTFT

Define the partial sum of $X(\mathrm{e}^{\mathrm{j}\Omega})$

$$X_N(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=-N}^{N} x[k]\mathrm{e}^{-\mathrm{j}\Omega k}$$

If $\sum_{k=-\infty}^{\infty} |x[k]| < \infty$ is absolutely summable

Then $\lim_{N\to\infty} \left| X(\mathrm{e}^{\mathrm{j}\Omega}) - X_N(\mathrm{e}^{\mathrm{j}\Omega}) \right| = 0$

If the sequence satisfies absolute summability, then the sequence has DTFT. (Sufficient condition)

Spectrum of discrete aperiodic signals

Features of $X(\mathrm{e}^{\mathrm{j}\Omega})$:

(1) $X(\mathrm{e}^{\mathrm{j}\Omega})$ is a continuous function of $\boldsymbol{\Omega}$.

(2) $X(\mathrm{e}^{\mathrm{j}\Omega})$ is a periodic function with a period of 2π.

$$X[\mathrm{e}^{\mathrm{j}(\Omega+2\pi)}] = \sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}(\Omega+2\pi)k}$$

$$= \sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}\Omega k} \cdot \mathrm{e}^{-\mathrm{j}2\pi k} = X(\mathrm{e}^{\mathrm{j}\Omega})$$

Spectrum of discrete aperiodic signals

Representation of sequence spectrum

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \underbrace{\left|X(\mathrm{e}^{\mathrm{j}\Omega})\right|}_{\text{Amplitude spectrum}} \mathrm{e}^{\mathrm{j}\underbrace{\varphi(\Omega)}_{\text{Phase spectrum}}}$$

The principal value interval of phase spectrum $\varphi(\Omega)$ is

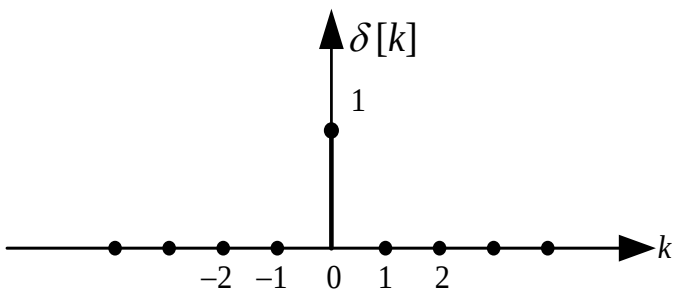
$$-\pi < \varphi(\Omega) \leq \pi$$

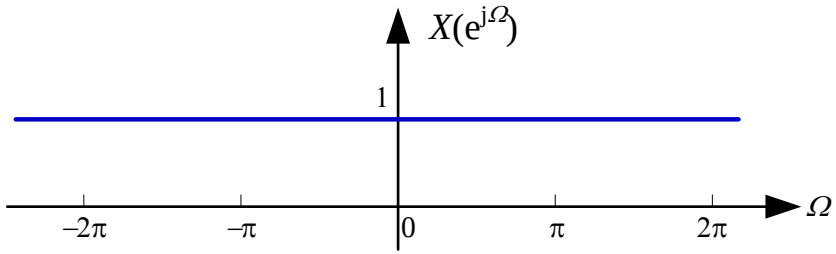
Spectrum of discrete aperiodic signals

Example: Find the frequency spectrum of unit pulse sequence $\delta[k]$.

Solution:

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=-\infty}^{\infty} \delta[k]\mathrm{e}^{-\mathrm{j}\Omega k} = 1$$





Spectrum of discrete aperiodic signals

Example: Find the frequency spectrum of the sequence $x[k]=\alpha^k u[k]$.

Solution:

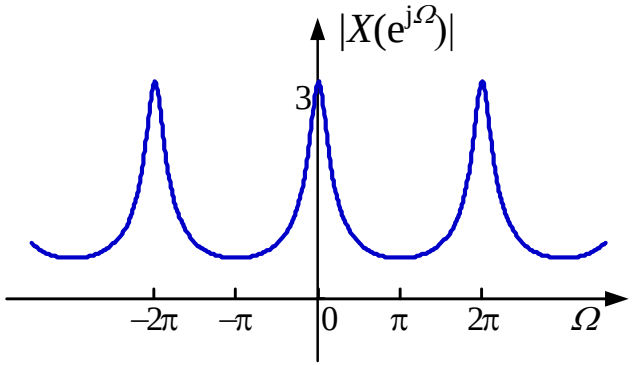
$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=0}^{\infty} \alpha^k \mathrm{e}^{-\mathrm{j}\Omega k} = \sum_{k=0}^{\infty} \left(\alpha \mathrm{e}^{-\mathrm{j}\Omega}\right)^k$$

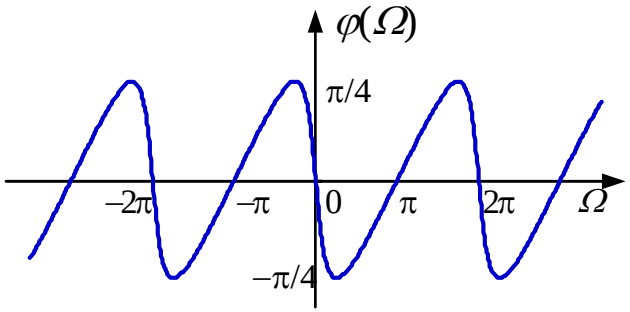
When |a|>1, Absolute summability is not satisfied, and the DTFT of the sequence does not exist.

When |a|<1,

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{1-\alpha \mathrm{e}^{-\mathrm{j}\Omega}}$$

α=0.7





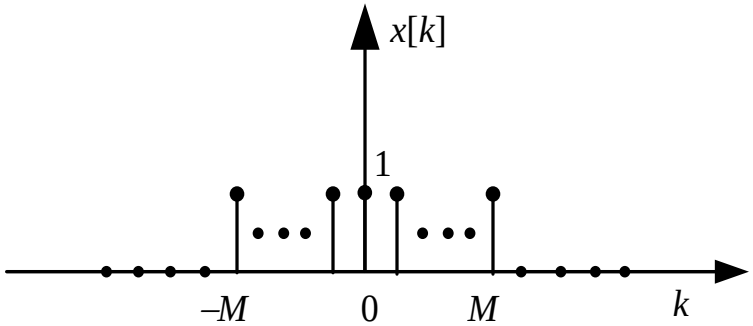
Spectrum of discrete aperiodic signals

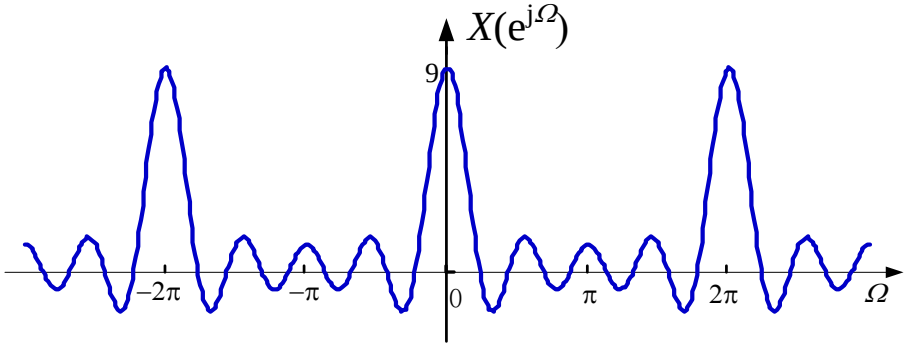
Example: Find the spectrum of a rectangular sequence $x[k]$ with a width of 2M+1.

Solution:

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}\Omega k} = \sum_{k=-M}^{M} \left(\mathrm{e}^{-\mathrm{j}\Omega}\right)^k$$

$$= \frac{\mathrm{e}^{\mathrm{j}M\Omega}\left[1-\mathrm{e}^{-\mathrm{j}(2M+1)\Omega}\right]}{1-\mathrm{e}^{-\mathrm{j}\Omega}} = \frac{\sin\left[\left(M+1/2\right)\Omega\right]}{\sin(\Omega/2)}$$





Spectrum of rectangular sequence when $M=4$

Properties of discrete aperiodic sequence DTFT

- linear characteristic
- Symmetry characteristic
- Displacement characteristic
- Convolution characteristic
- Differential characteristic
- Parseval Theorem

Properties of discrete aperiodic sequence DTFT

1. Linear characteristics

if

$$x_1[k] \xleftrightarrow{\text{DTFT}} X_1(\mathrm{e}^{\mathrm{j}\Omega})$$

$$x_2[k] \xleftrightarrow{\text{DTFT}} X_2(\mathrm{e}^{\mathrm{j}\Omega})$$

then

$$ax_1[k] + bx_2[k] \xleftrightarrow{\text{DTFT}} aX_1(\mathrm{e}^{\mathrm{j}\Omega}) + bX_2(\mathrm{e}^{\mathrm{j}\Omega})$$

Properties of discrete aperiodic sequence DTFT

2. Symmetry characteristics

$$x^*[k] \xleftrightarrow{\text{DTFT}} X^*(\mathrm{e}^{-\mathrm{j}\Omega})$$

$$x^*[-k] \xleftrightarrow{\text{DTFT}} X^*(\mathrm{e}^{\mathrm{j}\Omega})$$

When $x[k]$ is a real sequence

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = X^*(\mathrm{e}^{-\mathrm{j}\Omega})$$

Amplitude and phase

$$\left|X(\mathrm{e}^{\mathrm{j}\Omega})\right| = \left|X(\mathrm{e}^{-\mathrm{j}\Omega})\right|$$

$$\varphi(\Omega) = -\varphi(-\Omega)$$

Real part and imaginary part

$$X_{\mathrm{R}}(\mathrm{e}^{\mathrm{j}\Omega}) = X_{\mathrm{R}}(\mathrm{e}^{-\mathrm{j}\Omega})$$

$$X_{\mathrm{I}}(\mathrm{e}^{\mathrm{j}\Omega}) = -X_{\mathrm{I}}(\mathrm{e}^{-\mathrm{j}\Omega})$$

Properties of discrete aperiodic sequence DTFT

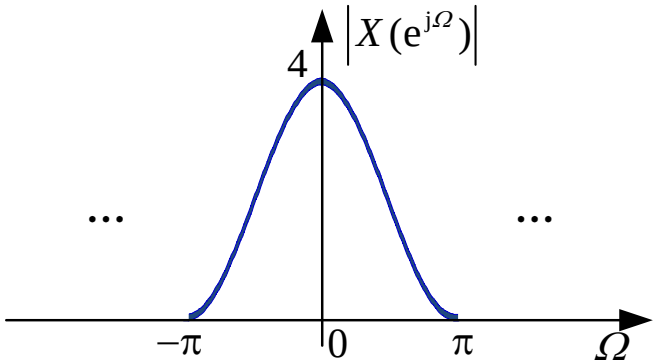
Example: Find the amplitude spectrum and phase spectrum of the sequence $x[k]=\{1,2,1;\ k=0,1,2\}$.

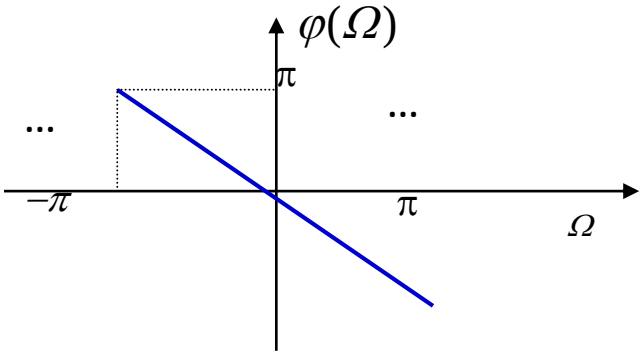
Solution:

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = 1 + 2\mathrm{e}^{-\mathrm{j}\Omega} + \mathrm{e}^{-\mathrm{j}2\Omega} = (1+\mathrm{e}^{-\mathrm{j}\Omega})^2$$

$$= 4\cos^2\left(\Omega/2\right)\cdot \mathrm{e}^{-\mathrm{j}\Omega}$$

$$\left|X(\mathrm{e}^{\mathrm{j}\Omega})\right| = 4\cos^2\left(\Omega/2\right) \qquad \varphi(\Omega) = -\Omega$$





Properties of discrete aperiodic sequence DTFT

3. Displacement characteristics

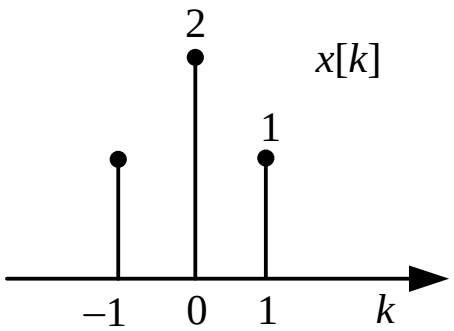
(a) Displacement characteristics in time domain

$$x[k-n] \xleftrightarrow{\text{DTFT}} X(\mathrm{e}^{\mathrm{j}\Omega})\mathrm{e}^{-\mathrm{j}\Omega n}$$

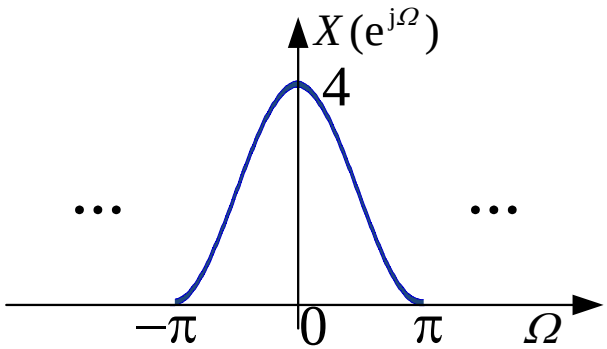
The displacement of a sequence in time domain corresponds to its phase shift in frequency domain.

Properties of discrete aperiodic sequence DTFT

Example: Given $x[k]$ as shown in the figure, find the frequency spectrum of $Y[k] = X[k\text{-}1]$.

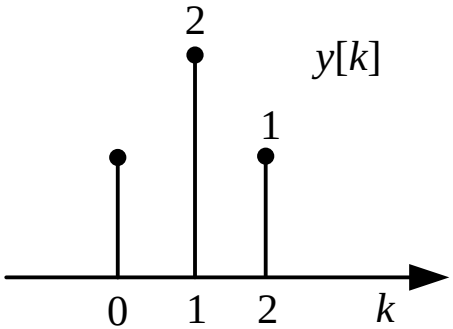


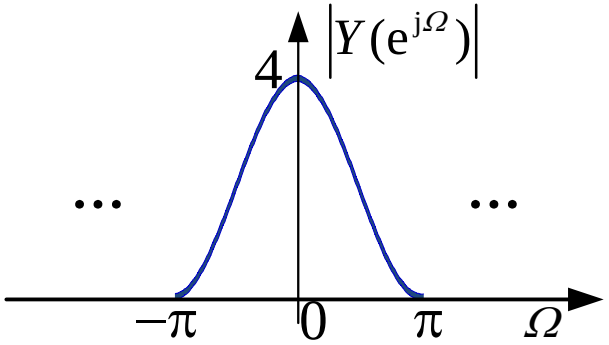
$$X(\mathrm{e}^{\mathrm{j}\Omega}) = 2(1+\cos\Omega)$$

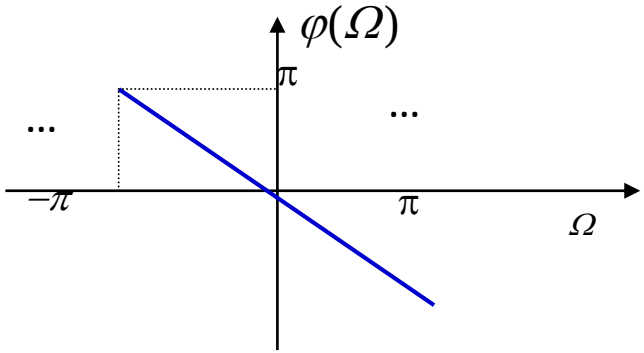


Solution:

$$Y(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{e}^{-\mathrm{j}\Omega} X(\mathrm{e}^{\mathrm{j}\Omega})$$







Properties of discrete aperiodic sequence DTFT

3. Displacement characteristics

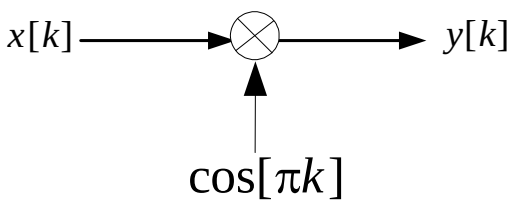
(b) Displacement characteristics in frequency domain

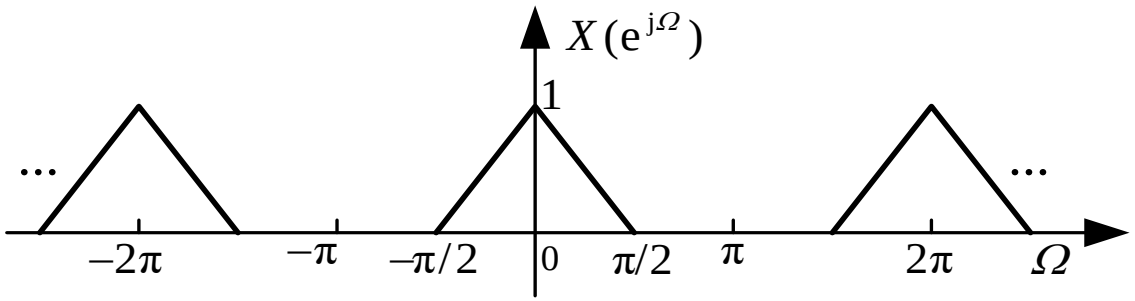
$$\mathrm{e}^{\mathrm{j}\Omega_0 k} x[k] \xleftrightarrow{\text{DTFT}} X(\mathrm{e}^{\mathrm{j}(\Omega-\Omega_0)})$$

The phase shift of a sequence in time domain corresponds to its frequency shift in frequency domain.

Properties of discrete aperiodic sequence DTFT

Example: Given the spectrum of $x[k]$ as shown in the figure, find the spectrum of $y[k]$.



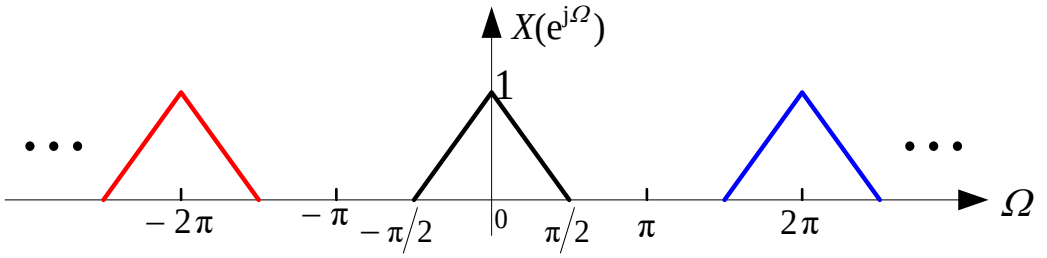


Solution: $$y[k] = x[k]\cos\left[\pi k\right] = x[k](\mathrm{e}^{\mathrm{j}\pi k} + \mathrm{e}^{-\mathrm{j}\pi k})/2$$

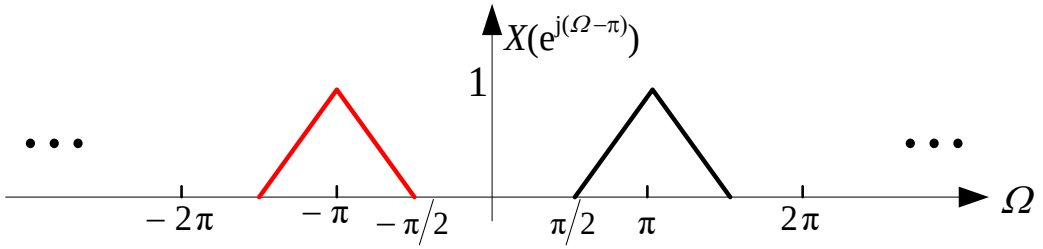
$$Y(\mathrm{e}^{\mathrm{j}\Omega}) = \left\{X(\mathrm{e}^{\mathrm{j}(\Omega-\pi)}) + X(\mathrm{e}^{\mathrm{j}(\Omega+\pi)})\right\}/2$$

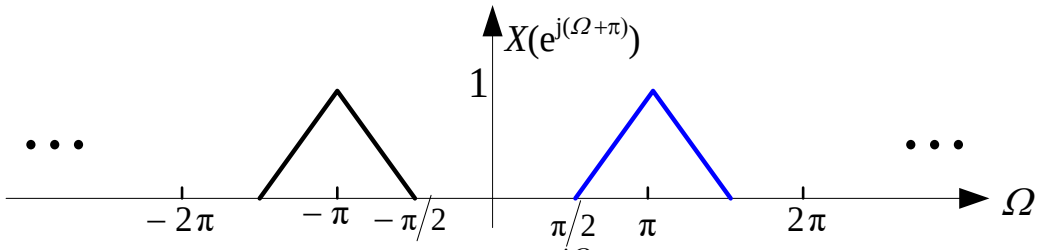
Properties of discrete aperiodic sequence DTFT

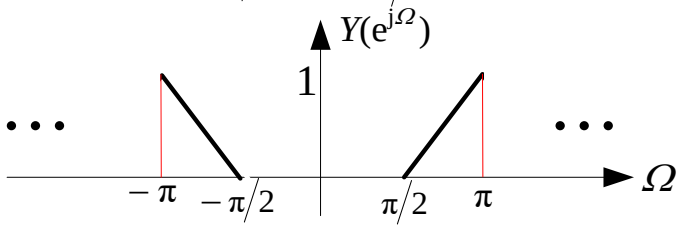
Example: Given the spectrum of $x[k]$ as shown in the figure, find the spectrum of $y[k]$.



Solution:







Properties of discrete aperiodic sequence DTFT

4. Convolution characteristics

(a) Convolution characteristics in time domain

$$x[k] * h[k] \xleftrightarrow{\text{DTFT}} X(\mathrm{e}^{\mathrm{j}\Omega})H(\mathrm{e}^{\mathrm{j}\Omega})$$

The convolution of sequence in time domain corresponds to the product of frequency domain.

4. Convolution characteristics

(a) Convolution characteristics in frequency domain

$$x[k]h[k] \xleftrightarrow{\text{DTFT}} \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\mathrm{e}^{\mathrm{j}\theta})H(\mathrm{e}^{\mathrm{j}(\Omega-\theta)})\mathrm{d}\theta$$

The product of sequence in time domain corresponds to convolution in frequency domain.

Properties of discrete aperiodic sequence DTFT

5. Frequency domain differential characteristics

$$kx[k] \xleftrightarrow{\text{DTFT}} \mathrm{j}\frac{\mathrm{d}X(\mathrm{e}^{\mathrm{j}\Omega})}{\mathrm{d}\Omega}$$

Properties of discrete aperiodic sequence DTFT

6. Parseval theorem

$$\sum_{k=-\infty}^{\infty} |x[k]|^2 = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} \left|X(\mathrm{e}^{\mathrm{j}\Omega})\right|^2 \mathrm{d}\Omega$$

The energy in the time domain of the sequence is equal to the energy in the frequency domain.

Prove:

$$\sum_{k=-\infty}^{\infty} |x[k]|^2 = \sum_{k=-\infty}^{\infty} x[k]x^*[k] = \sum_{k=-\infty}^{\infty} x[k]\left\{\frac{1}{2\pi}\int_{\langle 2\pi\rangle} X^*(\mathrm{e}^{\mathrm{j}\Omega})\mathrm{e}^{-\mathrm{j}\Omega k}\,\mathrm{d}\Omega\right\}$$

$$= \frac{1}{2\pi}\int_{\langle 2\pi\rangle} X^*(\mathrm{e}^{\mathrm{j}\Omega})\left\{\sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}\Omega k}\right\}\mathrm{d}\Omega = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} X^*(\mathrm{e}^{\mathrm{j}\Omega})X(\mathrm{e}^{\mathrm{j}\Omega})\mathrm{d}\Omega$$

$$= \frac{1}{2\pi}\int_{\langle 2\pi\rangle} \left|X(\mathrm{e}^{\mathrm{j}\Omega})\right|^2 \mathrm{d}\Omega$$

Properties of discrete aperiodic sequence DTFT

$x[k]$ is known to be a finitely long sequence and $x[k]=\{\overset{\downarrow}{1},2,3,4\}$. Try to directly determines the value of the following expression without calculating the spectrum $X(\mathrm{e}^{\mathrm{j}\Omega})$ of $x[k]$.

Solution:

(1) $$X(\mathrm{e}^{\mathrm{j}0})=\sum_{k=0}^{3}x[k]\cdot\mathrm{e}^{-\mathrm{j}k0}=\sum_{k=0}^{3}x[k]=1+2+3+4=10$$

(2) $$X(\mathrm{e}^{\mathrm{j}\pi})=\sum_{k=0}^{3}x[k]\cdot\mathrm{e}^{-\mathrm{j}k\pi}=\sum_{k=0}^{3}x[k]\cdot(-1)^{k}=1-2+3-4=-2$$

(3) $$\int_{-\pi}^{\pi}X(\mathrm{e}^{\mathrm{j}\Omega})\,\mathrm{d}\Omega=2\pi x[0]=2\pi$$

(4) $$\int_{-\pi}^{\pi}\left|X(\mathrm{e}^{\mathrm{j}\Omega})\right|^{2}\mathrm{d}\Omega=2\pi\sum_{k=0}^{3}\left|x[k]\right|^{2}=60\pi$$

Correspondence between time domain and frequency domain of four kinds of signals

The theoretical basis of signal frequency domain analysis is to represent the signal as a linear combination of sinusoidal (imaginary exponential) signals.

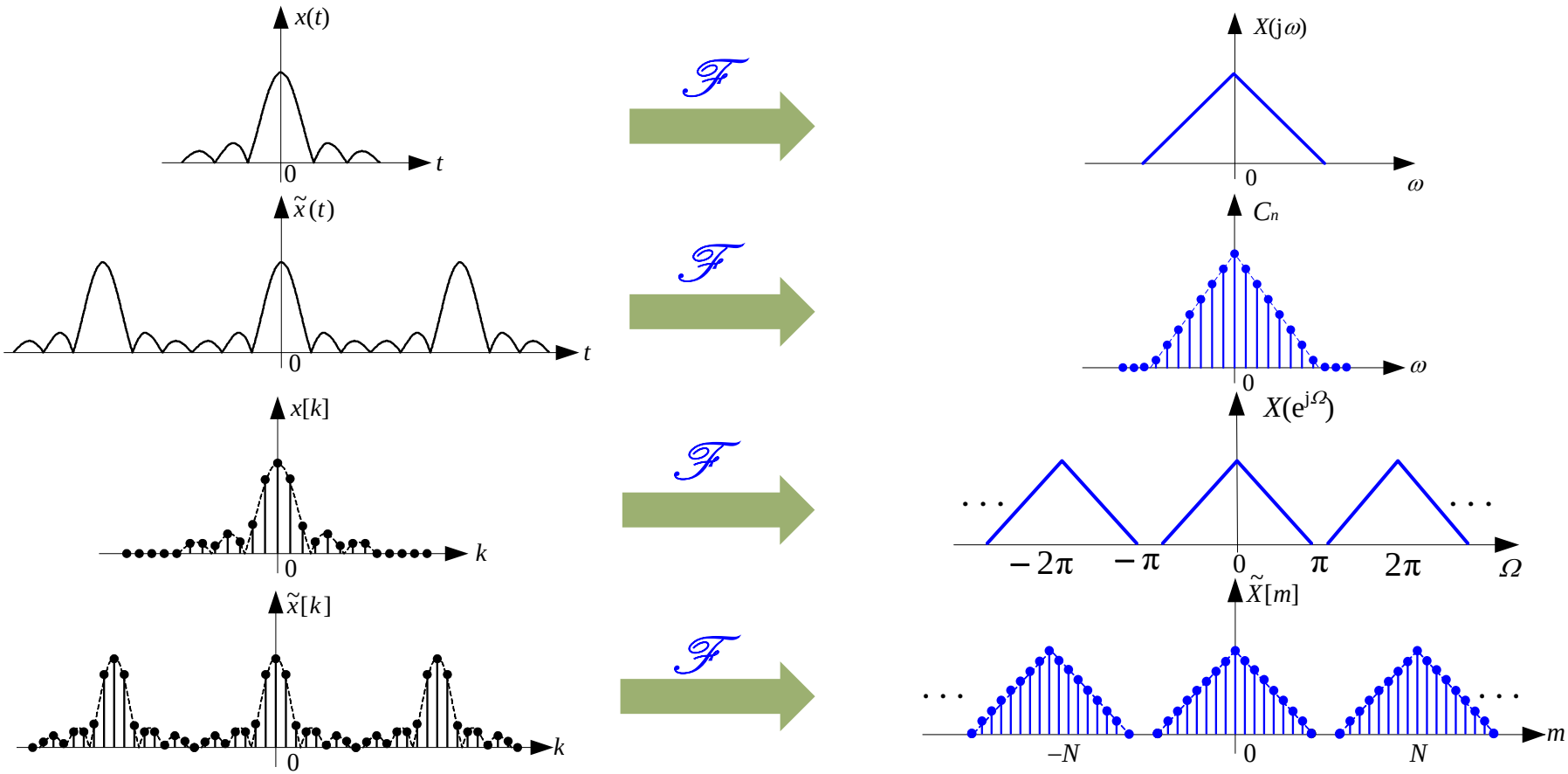
$$\tilde{x}(t)=\sum_{n=-\infty}^{\infty} C_n \cdot \mathrm{e}^{\mathrm{j}n\omega_0 t} \qquad C_n=\frac{1}{T_0}\int_0^{T_0}\tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

$$x(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty} X(\mathrm{j}\omega)\cdot \mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega \qquad X(\mathrm{j}\omega)=\int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$\tilde{x}[k]=\frac{1}{N}\sum_{m=0}^{N-1}\tilde{X}[m]\cdot \mathrm{e}^{\mathrm{j}\Omega_m k} \qquad \tilde{X}[m]=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}$$

$$x[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi} X(\mathrm{e}^{\mathrm{j}\Omega})\cdot \mathrm{e}^{\mathrm{j}\Omega k}\mathrm{d}\Omega \qquad X(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}\Omega k}$$

Correspondence between time domain and frequency domain of four kinds of signals



Correspondence between time domain and frequency domain of four kinds of signals

Continuous periodic signal
Continuous aperiodic signal

$$\tilde{x}(t) \Leftrightarrow C_n$$

Discrete aperiodic spectrum

$$x(t) \Leftrightarrow X(\mathrm{j}\omega)$$

Continuous aperiodic spectrum

Discrete periodic signal
Discrete aperiodic signal

$$\tilde{x}[k] \Leftrightarrow \tilde{X}[m]$$

Discrete periodic spectrum

$$x[k] \Leftrightarrow X(\mathrm{e}^{\mathrm{j}\Omega})$$

Continuous periodic spectrum

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!